

Astitva Gaur

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OBJECTIVE

B.Tech IT student at GGSIPU with hands-on experience in Python, FastAPI, and GenAI frameworks (LangChain, RAG). Co-author of a published paper on LLM memory management. Seeking AI/ML or backend engineering roles to apply and grow my skills in building intelligent systems.

EDUCATION

Guru Gobind Singh Indraprastha University *New Delhi, India*

B.Tech – Information Technology 2027

- CGPA: **8.59/10** (till 6th Semester)

TECHNICAL SKILLS

Languages: Python, C++ (academic)

Frameworks & Libraries: FastAPI, Flask, NumPy, Pandas, Matplotlib, Seaborn, Streamlit, Pydantic

Databases: MySQL, SQLite

Tools & Platforms: Git, GitHub, Swagger UI, JWT Authentication, REST APIs, CRUD Application Development

AI / GenAI: LangChain, RAG pipelines, Gemini API, LLM integration

PROJECTS

FloatChat – RAG-based Ocean Data Research Chatbot

Tech Stack: Python, FastAPI, LangChain, Gemini API, NetCDF, Pandas, Streamlit

- Architected **backend API routes and rate-limiting middleware** using FastAPI, ensuring stable multi-user access for oceanographic researchers.
- Built a **Retrieval-Augmented Generation (RAG) pipeline** integrating Gemini API to enable intelligent querying over complex **NetCDF ocean datasets**.
- Implemented **document ingestion and vector retrieval** workflows allowing researchers to extract insights from large-scale climate and ocean data files.
- Designed **conversational context management** for multi-turn dialogue, improving query resolution accuracy for domain-specific scientific questions.

AQI & Weather Correlation Analysis – Data Visualization Dashboard

Tech Stack: Python, Pandas, NumPy, Matplotlib, Seaborn, Streamlit

- Performed **exploratory data analysis (EDA)** on Air Quality Index (AQI) and weather datasets to uncover correlations between pollutants and meteorological variables.
- Built interactive **Streamlit dashboards** with Matplotlib and Seaborn visualizations to present coefficient relationships and trend patterns clearly.
- Applied **statistical correlation techniques** including heatmaps and regression plots to identify key environmental drivers affecting AQI levels.

How AI Works – Interactive AI Explainer Website [Live](#) | [GitHub](#)

Tech Stack: HTML5, CSS3, Vanilla JavaScript (ES6+), GSAP ScrollTrigger, Spline 3D

- Built a **cinematic, scroll-driven web experience** that visually explains core AI concepts — neural networks, tokenization, transformers, and backpropagation — across 6 interactive chapters.
- Integrated **Spline 3D runtime** for real-time 3D model rendering and **GSAP ScrollTrigger** for surgical scroll-based animation choreography.
- Implemented an **interactive probability scoring module** and live neural network visualization that responds to user interaction, demonstrating AI decision-making mechanics.
- Deployed on **Vercel** with a cyberpunk-inspired UI featuring glassmorphism, CSS noise textures, and GPU-accelerated animations.

RESEARCH & PUBLICATIONS

Addressing Memory Limitations in LLM-based AI Agents

Co-author | Published on ResearchGate & Zenodo | [View Paper](#)

- Co-authored a research paper investigating **memory management strategies** for LLM-based AI agents, addressing context window limitations and long-term knowledge retention.
- Explored architectural approaches including **external memory stores, retrieval-augmented memory, and agent state management** to enhance agent performance over extended interactions.

HACKATHONS & COMPETITIONS

- **T-Hacks – 3rd Place** | Competed among 400+ teams; recognized for innovative system design concept in anomaly detection domain.
- **Vespera (USAR, IPU) – Top 10** | Ranked in top 10 among 400+ participating teams.
- **Hacknovate – Top 10** | Achieved top 10 finish out of 150–200 competing teams.
- **GeeksforGeeks Hackathon** | Qualified multiple rounds, placing among top 20 teams nationally.

CERTIFICATIONS & COURSES

- **Python Programming** – Codedex (gamified, project-based learning platform)
- **Data Science with Python** – Udemy (Krish Naik) – In Progress

INTERESTS

Artificial Intelligence & Generative AI | Guitar & Music | Volleyball